Poisson regression

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Previously: Activity

Use the provided packet to answer the following questions:

- Does the Poisson regression model appear to be a good fit?
- If not, which piece of the model do you think is mis-specified?

A goodness-of-fit test

H_0: model is a good fit to the data

H_A: model is <u>not</u> a good fit to the data

<u>Idea</u>: find a test statistic which behaves a certain way under H_0

If ϕ is known and H_0 is true

$$\frac{1}{n-p}\sum_{i=1}^{n} d(y_i,\hat{\mu}_i) = \frac{1}{n-p} D(y,\hat{\mu}) \approx \phi$$

$$\frac{1}{n-p}\sum_{i=1}^{n} \frac{(y_i-\hat{\mu}_i)^2}{V(\hat{\mu}_i)} \approx \phi$$

Possible test statistics:

$$\frac{D(y,\hat{\mu})}{\phi}$$

$$\frac{1}{\phi}\sum_{i=1}^{n} \frac{(y_i-\hat{\mu}_i)^2}{V(\hat{\mu}_i)}$$

Null distribution??

Parametric bootstrapping

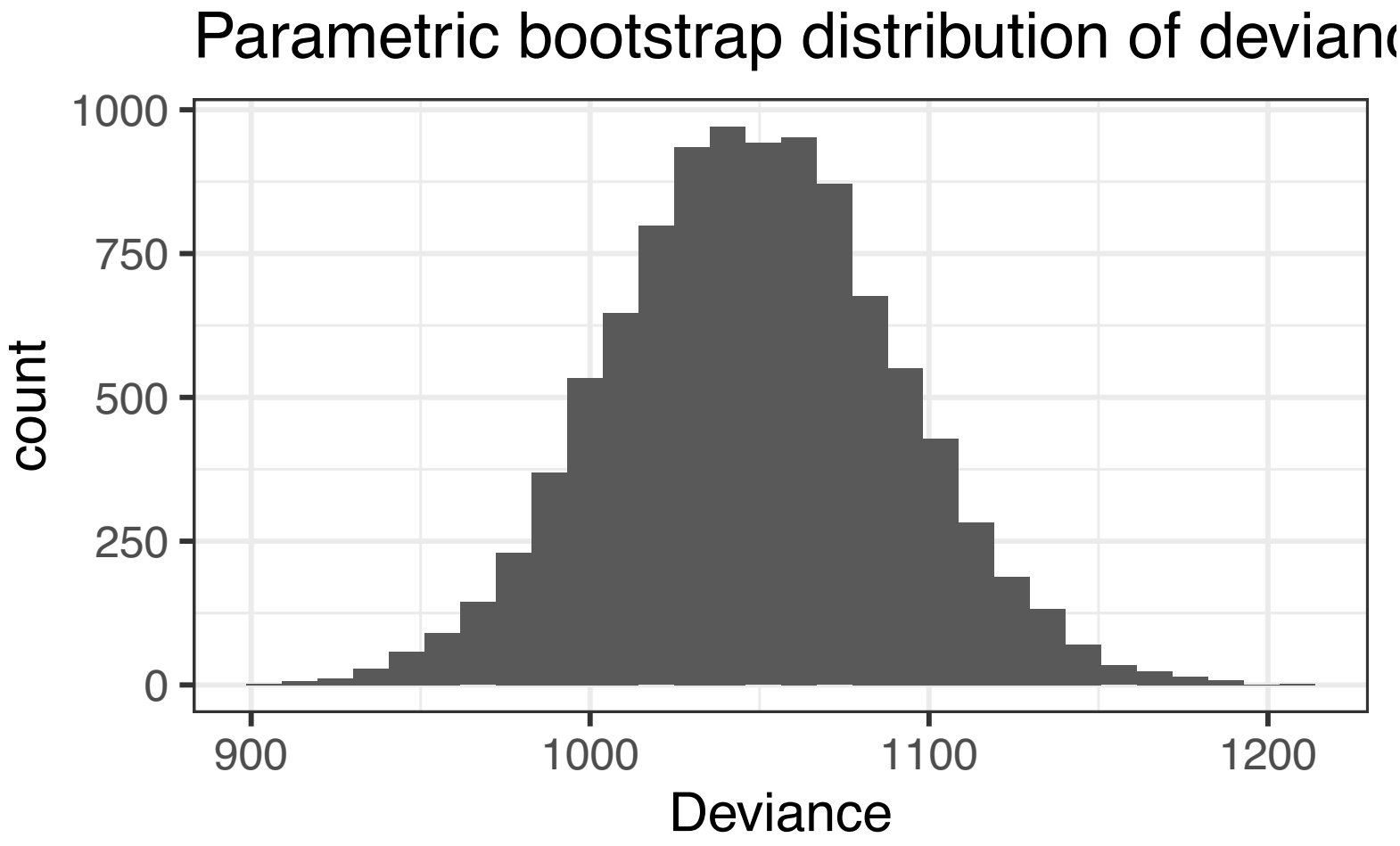
- Fit model on original data $\rightarrow \hat{\mu}_1, \ldots, \hat{\mu}_n$
- Simulate new Y_i^* from fitted model

(so H_0 must be true for simulated data)

$$Y_i^* \sim \text{Poisson}(\hat{\mu}_i)$$

- Re-fit model on $(x_1, Y_1^*), \ldots, (x_n, Y_n^*)$
- Calculate test statistic on model with simulated data
- Repeat,

Parametric bootstrapping




```
m_art$deviance
```

$= D(y, \hat{\mu}) = \frac{D(y, \hat{\mu})}{\phi}$ $(\phi = 1)$

```
## [1] 1661.067
```

very unusual if H_0 is true!

Approximate null distribution

If ϕ is known:

- If saddlepoint approximation is good:

$$\frac{D(y, \hat{\mu})}{\phi} = \frac{1}{\phi} \sum_{i=1}^{n} d(y_i, \hat{\mu}_i) \approx \chi^2_{n-p}$$

- If ϕ is close to 0:

$$\frac{1}{\phi} \sum_{i=1}^{n} \frac{(y_i - \mu_i)^2}{V(\hat{\mu}_i)} \approx \chi^2_{n-p}$$

If saddlepoint approx. is good:

$$f(y; \mu, \phi) \approx \frac{1}{\sqrt{2\pi\phi V(y)}} \exp\left\{ - \frac{d(y, \mu)}{2\phi} \right\}$$

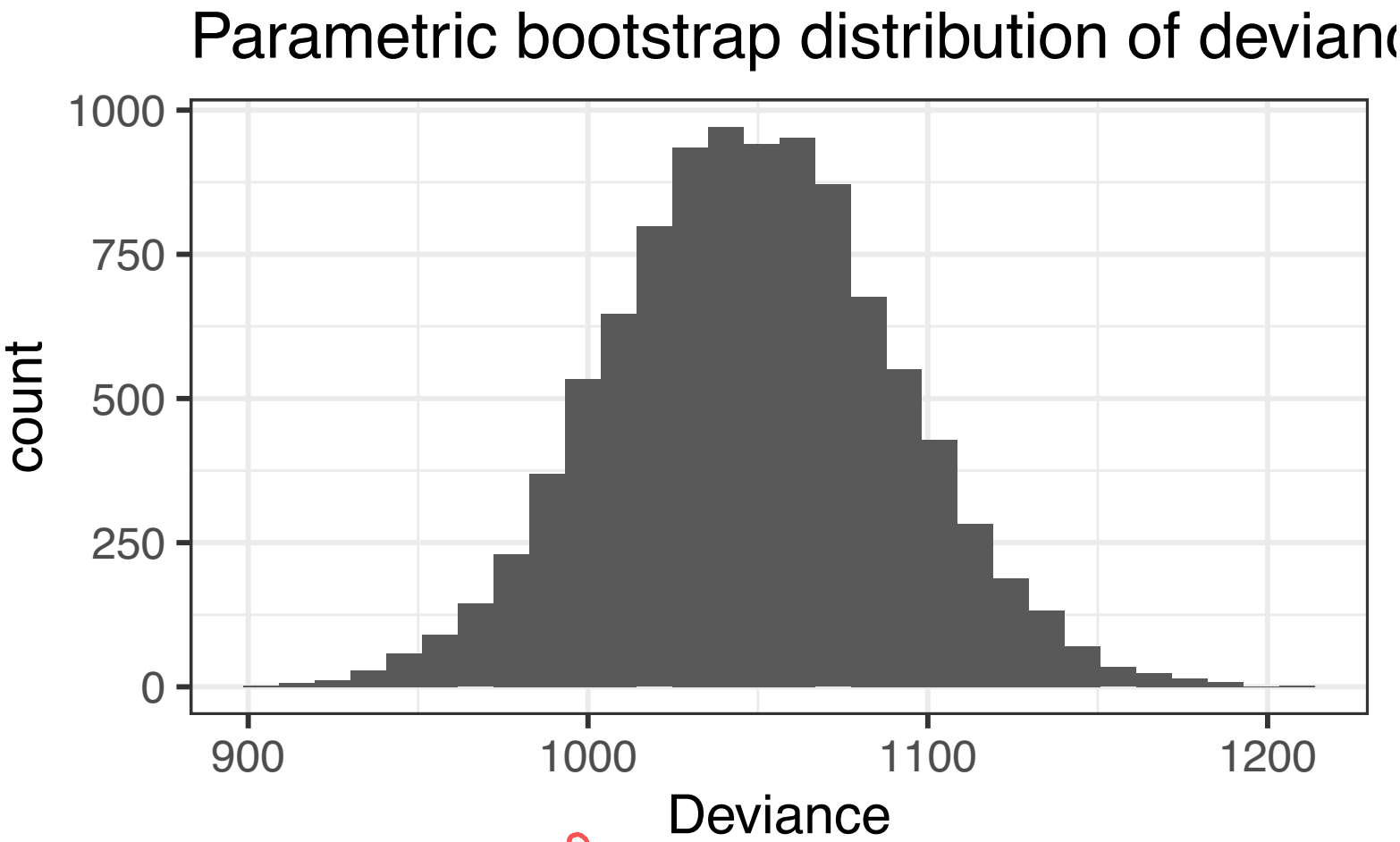
$$\Rightarrow \frac{d(y, \mu)}{\phi} \approx \chi^2_1 \Rightarrow \frac{1}{\phi} \sum_{i=1}^{n} d(y_i, \hat{\mu}_i) \approx \chi^2_{n-p}$$

Pf: Dunn & Smyth Section 5.4.3

Intuition: $N(\mu, \sigma^2)$: $d(y, \mu) = (y - \mu)^2$

$Z = \frac{Y - \mu}{\sigma} \sim N(0, 1)$ $\left(\frac{Y - \mu}{\sigma}\right)^2 = Z^2 \sim \chi^2_1$

Parametric bootstrapping



This isn't quite a χ^2_{911} distribution. Why not?

n-p

Because saddle point approx. isn't quite accurate

Dunn & Smyth suggest $y \geq 3$ for approx. to be good

Another dataset

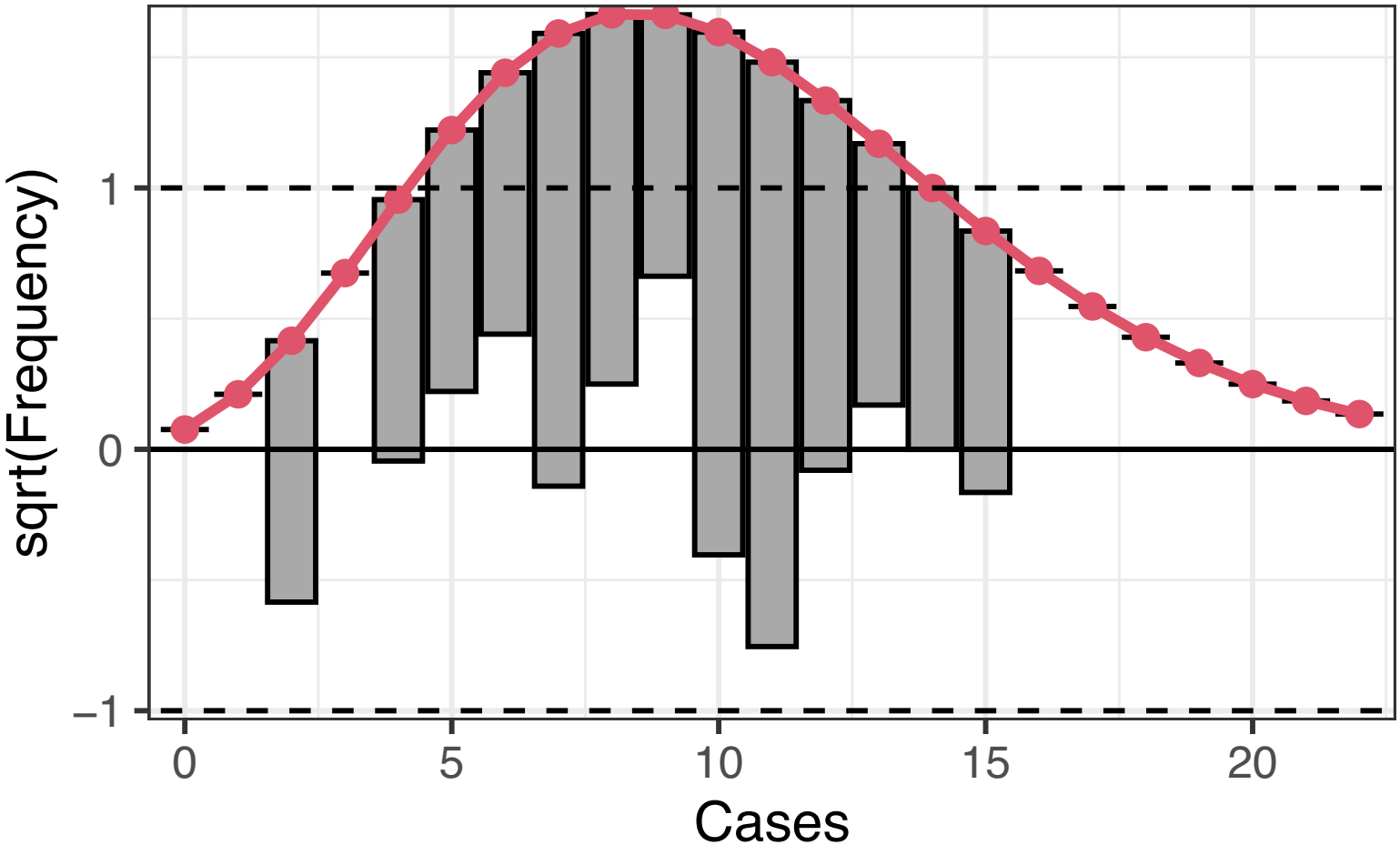
(Each row represents one age group in one city)

Data on the number of lung cancer cases in different Danish cities.
Poisson regression model:

```
m_lung <- glm(Cases ~ City + Age,
              family = poisson, data = danishlc)
summary(m_lung)$coefficients
```

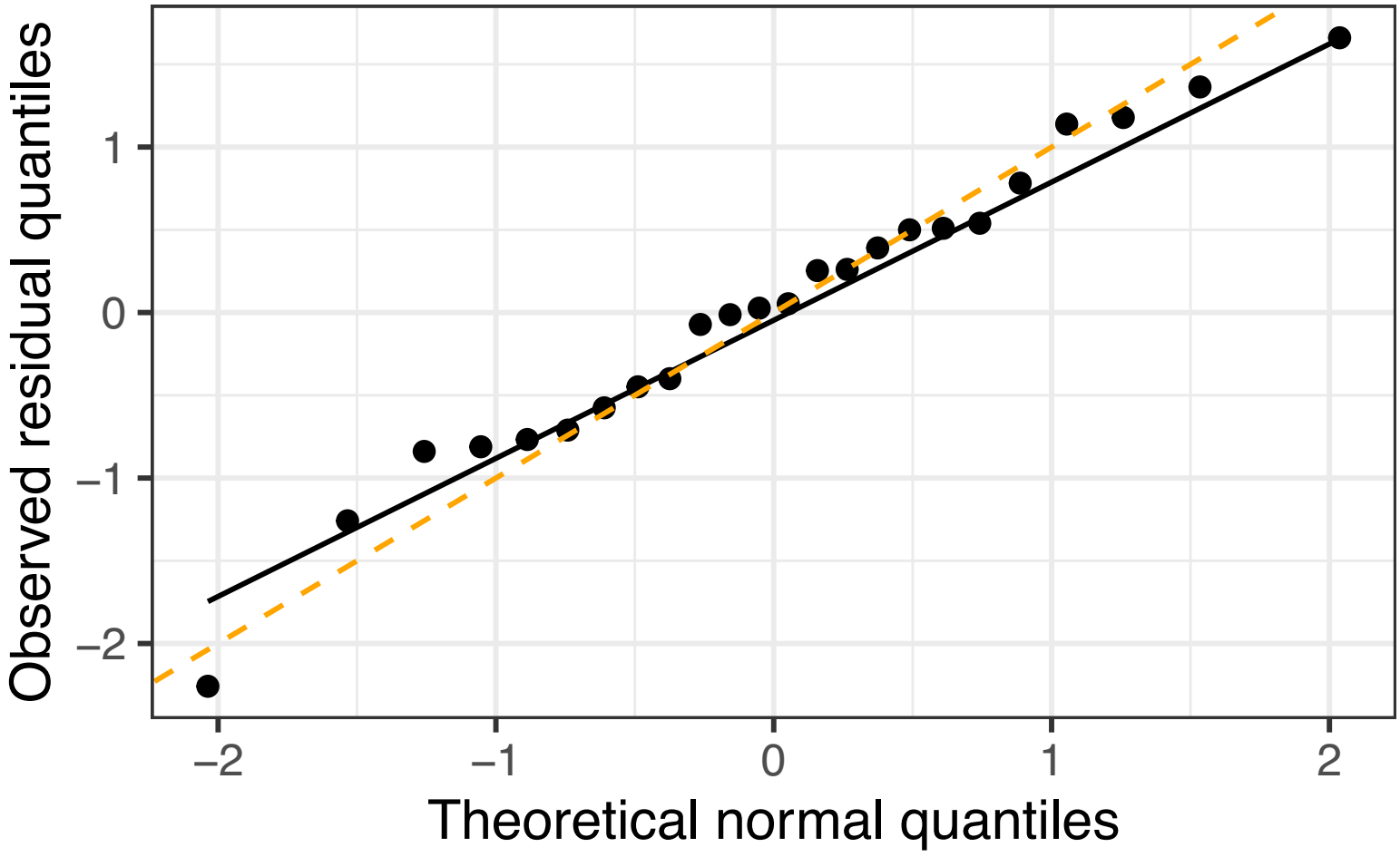
```
##                  Estimate Std. Error    z value      Pr(>|z
## (Intercept)  2.18122424  0.2083717 10.4679470 1.212440e-
## CityHorsens -0.09844007  0.1812909 -0.5429952 5.871331e-
## CityKolding -0.22705745  0.1877041 -1.2096561 2.264109e-
## CityVejle   -0.22705745  0.1877041 -1.2096561 2.264109e-
## Age40-54     0.06252036  0.2501222  0.2499593 8.026188e-
## Age55-59     0.03174870  0.2520081  0.1259829 8.997455e-
## Age60-64     0.32721291  0.2356138  1.3887678 1.649034e-
## Age65-69     0.37267529  0.2334101  1.5966543 1.103428e-
## Age70-74     0.25489225  0.2392866  1.0652175 2.867775e-
```

Some diagnostics



only 24 observations

Some diagnostics



Some diagnostics

```
table(danishlc$Cases)
```

almost all $y \geq 3$ => saddle point approx. probably ok

```
##
##  2  4  5  6  7  8  9 10 11 12 13 14 15
##  1  1  1  1  3  2  1  4  5  2  1  1  1
```

```
m_lung$deviance
```

```
## [1] 20.67302
```

```
pchisq(m_lung$deviance, m_lung$df.residual, lower.tail=F)
```

```
## [1] 0.1476057
```

Model interpretation

```
##                  Estimate Std. Error    z value     Pr(>|z
## (Intercept)  2.18122424  0.2083717 10.4679470 1.212440e-
## CityHorsens -0.09844007  0.1812909 -0.5429952 5.871331e-
## CityKolding -0.22705745  0.1877041 -1.2096561 2.264109e-
## CityVejle   -0.22705745  0.1877041 -1.2096561 2.264109e-
## Age40-54     0.06252036  0.2501222  0.2499593 8.026188e-
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## Age70-74     0.25489225  0.2392866  1.0652175 2.867775e-
```

Are there any additional variables we would want to add to the model?

Offsets

We can account for population size by including an **offset** in the model:

population for that age group in that city

$$\log(\mu_i) = \beta_0 + \beta_1 \text{Horsens}_i + \cdots + \beta_8 \text{Age70-74}_i + \log(\text{Population}_i)$$

$$\Rightarrow \log\left(\frac{\mu_i}{\text{Population}}\right) = \beta_0 + \beta_1 \text{Horsens}_i + \cdots + \beta_8 \text{Age70-74}_i$$

rate of lung cancer

(average # of cases per unit of population)

$\Rightarrow$ can now interpret βs in terms of rates rather than counts

But response is still Poisson

Fitting a model with offsets

```
m_lung <- glm(Cases ~ City + Age, offset = log(Pop),
              family = poisson, data = danishlc)
```

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-4.21241109	0.2081915	-20.2333461	4.980576e-91
CityHorsens	-0.33005999	0.1815033	-1.8184791	6.899094e-02
CityKolding	-0.37154616	0.1878063	-1.9783480	4.788946e-02
CityVejle	-0.27231769	0.1878534	-1.4496291	1.471620e-01
Age40-54	-1.41965342	0.2502707	-5.6724722	1.407514e-08
Age55-59	-0.31863942	0.2520492	-1.2641953	2.061599e-01
Age60-64	0.09895893	0.2356616	0.4199197	6.745441e-01
Age65-69	0.34805283	0.2334288	1.4910451	1.359497e-01
Age70-74	0.43720988	0.2392942	1.8270809	6.768761e-02

E.g. $\exp\{-0.3186\} \times 0.727$

Holding city fixed, the expected # of cases per 1000 people in the 55-59 bracket is about 27.3% lower than in >74 age bracket

Offsets

Offsets are useful when our counts come from groups of very different sizes (e.g., different cities). The offset lets us interpret model coefficients in terms of rates rather than average counts.

Question: Can you think of other examples of data in which the response is a count variable, and an offset would be useful?

offsets measure differences in exposure
different population size, land area,
lengths of time, # of opportunities, etc.

Articles data

- `art`: articles published by the student in last three years of Ph.D.
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program
- `ment`: articles published by their mentor in last three years

Question: Do you think we would want an offset for a Poisson model with this data?

No - no difference in exposure